

1. spendingAnalyzer.js

Purpose

Raw spending statistics.

It should answer:

- How much spent?
- How many transactions?
- Daily average?
- Weekly average?
- Monthly average?
- Largest expense?
- Smallest expense?
- Average transaction amount?

It should NOT care about budgets or habits.

Returns

```
{  
  totalSpent,  
  transactionCount,  
  averageTransaction,  
  dailyAverage,  
  weeklyAverage,  
  monthlyAverage,  
  largestExpense,  
  smallestExpense  
}
```

2. budgetAnalyzer.js

This is ONLY about budget.

Questions:

- Budget utilized %
- Remaining budget
- Overspending?
- Budget left

- Budget streak
- Days until exhaustion
- Projected overspend

Returns

```
{  
  budget,  
  spent,  
  remaining,  
  utilization,  
  status,  
  projectedEndOfMonth,  
  streak  
}
```

3. categoryAnalyzer.js

Anything category-related.

Questions

- Top category
- Least category
- Category percentages
- Spending distribution
- Biggest category increase
- Biggest category decrease

Returns

```
{  
  topCategory,  
  leastCategory,  
  categoryDistribution,  
  categoryGrowth,  
  biggestJump,  
  biggestDrop  
}
```

4. trendAnalyzer.js

This compares time periods.

Questions

- Compared to yesterday
- Compared to last week
- Compared to last month
- Compared to previous quarter

Returns

```
{  
  dailyTrend,  
  weeklyTrend,  
  monthlyTrend,  
  yearlyTrend,  
  spendingDirection  
}
```

5. habitAnalyzer.js

This is behavioral analysis.

Not financial health.

Questions

- Weekend vs weekday
- Night spending
- Morning spending
- Micro spending
- Impulse spending
- Subscription pattern
- Shopping frequency

Returns

```
{  
  weekdayAverage,  
  weekendAverage,  
  microSpending,  
}
```

```
    impulseScore,  
    frequentCategories,  
    shoppingPattern  
}
```

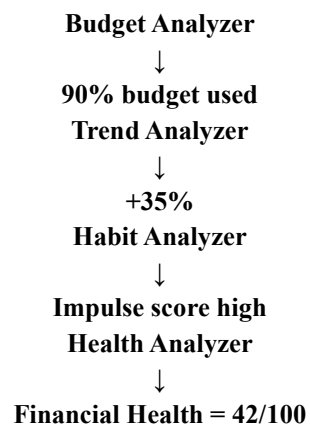
6. healthAnalyzer.js

This is the highest-level analyzer.

It DOES NOT calculate raw statistics.

Instead, it consumes outputs from all other analyzers.

Example



This analyzer combines everything.

Returns

```
{  
  stabilityScore,  
  healthScore,  
  riskLevel,  
  strengths,  
  weaknesses,  
  recommendations  
}
```

7. reportGenerator.js

This should never perform analysis.

Its job is only to merge outputs.

```
{  
  spending: ...,  
  budget: ...,  
  category: ...,  
  trend: ...,  
  habit: ...,  
  health: ...  
}
```

and maybe

```
{  
  generatedAt,  
  month,  
  report  
}
```

It acts as the assembler for your frontend or SIA.

analyticsEngine.generateReport()



fetchTransactions()

fetchBudgets()



spendingAnalyzer.analyze(transactions)

```
budgetAnalyzer.analyze({  
    budgets,  
    transactions  
})
```

categoryAnalyzer.analyze(transactions)

trendAnalyzer.analyze(transactions)

habitAnalyzer.analyze(transactions)

```
healthAnalyzer.analyze({  
    spending,  
    budget,  
    trend,  
    habit,  
    category  
})
```